

Aditya Antony Thomas

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AI/Data Engineer

AI and Data Engineer with a Master's in Data Science, a peer-reviewed publication in Results in Engineering, and hands-on experience building production Snowflake pipelines, SCD Type 2 implementations, and ML models for time series forecasting and NLP. Seeking to build robust data infrastructure and intelligent systems that drive measurable business outcomes.

Publications

- Aastha M. Sathe, Supraja R, Aditya Antony Thomas. Enhancing agricultural sustainability: Time series forecasting with ICEEMDAN-VMD-GRU for economic-resilience. *Results in Engineering*. Cited 7 times. In press.
<https://doi.org/10.1016/j.rineng.2025.106423>

Skills

Languages: Python, SQL

Frameworks/Libraries: TensorFlow, Scikit-Learn, VectorBT, Pandas, NumPy, Matplotlib, Seaborn, FastAPI, ReactJS

Tools/Platforms: Snowflake, AWS S3, Tableau, Selenium, Supabase

Experience/Internships

Data Engineer, tsworks – Bangalore

Dec 2025 – Ongoing

- Engineered an end-to-end data pipeline ingesting Shopify inventory and transactional data from AWS S3 external stages into Snowflake, independently delivering 62 of 90 tables and stored procedures.
- Designed and implemented SQL transformations and aggregation stored procedures to compute retail KPIs including sales summary, stockout items, stockout loss, and deadstock for a downstream BI dashboard.
- Rebuilt SCD Type 2 implementation from scratch, resolving critical data integrity issues present in the previous pipeline and ensuring accurate historical tracking of inventory and transactional records.
- Collaborated on the architecture of an internal agentic AI system, gaining exposure to LLM-based workflow design and multi-step agent development.

Data Analyst Intern, Ideas to Impact Foundation – Delhi

Aug 2025 – Dec 2025

- Validated 10,000+ MSME records sourced from third-party field agencies by cross-matching reported data against confirmed client records, identifying widespread data falsification.
- Detected fraudulent submissions by identifying temp email domain patterns across vendor data, leading to strict disciplinary action against a third-party agency.
- Wrote Python scripts to automate data splitting for Airtable uploads and to fuzzy-match incorrectly recorded field emails against confirmed client provided emails, improving data accuracy.
- Coordinated with an external vendor to implement an Odoo dashboard for streamlining the MSME data validation workflow, defining requirements and overseeing implementation.
- Reconciled discrepancies between Looker-reported metrics and internal reporting figures using Excel and Python, uncovering data handling issues in existing pipelines.

Quant Researcher Intern, FounderGraphy – Remote

Jan 2025 – Apr 2025

- Scraped and classified ~6,000 news articles across 50 companies over 2 months using ChatGPT API, enriching trading signal datasets for downstream analysis.
- Built a backtesting algorithm to optimize trading signals, evaluating strategy effectiveness using win rate and total return metrics.
- Automated daily Upstox API login using Selenium and designed an algorithm to retrieve option chain data, eliminating manual intervention.
- Implemented technical indicators (RSI, Moving Averages, MACD) and optimized strategies using heatmaps and

performance metrics.

- Applied hyperparameter tuning and backtesting frameworks to improve predictive accuracy and reduce overfitting.

Data Preprocessor Intern, KonectU – Remote

Jun 2024 – Aug 2024

- Built a Python preprocessing pipeline that cleaned and structured 20,000+ lines of raw text into training-ready format in under 3 minutes, enabling faster model iteration.

Projects

ListifyAI – Etsy Listing Optimizer | listify-ai-brown.vercel.app

May 2026

- Built and deployed a full-stack AI SaaS tool generating SEO-optimized Etsy listings using Groq API (Llama 3.3 70b), with a ReactJS + Vite frontend, FastAPI backend, Supabase PostgreSQL database, Google OAuth authentication, deployed on Vercel and Render.

Time Series Forecasting with CEEMDAN-VMD-LSTM for Vegetable Prices

Nov 2024

- Built a hybrid deep learning model combining CEEMDAN and VMD signal decompositions with an LSTM backbone to forecast vegetable prices from UK Government data.
- Achieved a Mean Absolute Percentage Error of 8%, validating the model's practical forecasting accuracy.

Time Table System and Assignment Manager

Jan 2024

- Built a full-stack web application with role-based access for teachers and students, supporting class enrolment, PDF assignment distribution, submission management, and grade tracking.

Participations/Achievements

Special Team Award

May 2026

Received for successfully creating and deploying an internal application to Shopify Marketplace.

Amazon ML Challenge

Aug 2024

Built an OCR and Regex-based pipeline to extract and classify text from product images.

Best Paper Presentation, Asian Business School International Conference (Finance Category)

Nov 2024

Presented on the impact of sentiment analysis on stock price prediction.

Education

Vellore Institute of Technology, AP, Masters in Data Science

Apr 2025

- CGPA: 8.45
- Coursework: Machine Learning, Deep Learning, Statistics, Data Visualization, DBMS

Loyola College, Chennai, Bachelors in Physics

2019 - 2022

- CGPA: 8.44

Certificates

Azure AI Fundamentals - Microsoft

Mar 2026

Foundations of AI and Machine Learning (Microsoft-Coursera)

Jul 2025